

Curriculum Vitae

Matthew Corbelli
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Education

PhD in Mathematics, June 2025

University of California, Davis

Dissertation: *Braided Tensor Categories Describing Anyons in Quantum Lattice Systems: Symmetries, Fermions, and the Fractional Quantum Hall Effect*

Master of Science in Mathematics, June 2021

University of California, Davis

Bachelor of Science in Mathematics (with a minor in Computer Science), June

2018 University of California, Davis

Research Experience

Fractional Quantum Hall Effect (FQHE):

- Collaborated with M. Fraas, S. Bachmann, and Y. Ogata to establish an upper bound for the Hall conductance denominator in infinite-volume lattice systems, linked to the number of inequivalent simple objects in the associated category.

Symmetry in Lattice Systems:

- Extended Ogata's version of a construction of a braided C^* -tensor category for a lattice system to incorporate requirements of compatibility with on-site symmetries and anticommuting Fermion-like operators.
- Modified Naaijken's (2022) theorem to show that imposing certain symmetry requirements leads to the sectors of product states system being classified by the Pontryagin dual of the symmetry group, instead of having a trivial sector theory.

Computational Complexity:

- Defined a new complexity class for multi-player team games (in collaboration with D. Tobias and S. Dutta) and demonstrated that this class contains $\Delta_2\text{EXP}$.

Publications

Sven Bachmann, Matthew Corbelli, Martin Fraas, and Yoshiko Ogata. *Tensor Category Describing Anyons in the Quantum Hall Effect and Quantization of Conductance*. *Reviews in Mathematical Physics* (2025), DOI: [10.1142/S0129055X24610075](https://doi.org/10.1142/S0129055X24610075). Also available as [arXiv:2410.04736](https://arxiv.org/abs/2410.04736).

Matthew Corbelli. *On Symmetry-Compatible Superselection Structures for Product States in 2D Quantum Spin Systems*. Preprint, [arXiv:2510.23790](https://arxiv.org/abs/2510.23790).

Appointments

- Graduate Teaching Assistant, University of California, Davis 2020–2025
- Graduate Research Assistant, University of California, Davis selected terms

Funding

- Departmental funding and internal fellowships, University of California, Davis
- Graduate research support in part from the National Science Foundation (DMS-2407290)

Teaching Experience

- Graduate Teaching Assistant, University of California, Davis 2020–2025
- Calculus I (Math 21A): Winter 2021, Fall 2023
- Calculus II (Math 21B): Winter 2021 (Lead TA), Fall 2021
- Multivariable Calculus and Vector Analysis (Math 21D): Winter 2023, Fall 2024
- Short Calculus Sequence (Math 16A/B/C): Spring 2020, Fall 2020, Spring 2021, Spring 2022, Winter 2025
- Precalculus (Math 12): Spring 2024
- Calculus for the Biological Sciences (Math 17C): Spring 2023
- Responsibilities included leading discussion sections, holding office hours, staffing a drop-in calculus help center, grading assignments, and proctoring exams.

Research Interests / Skills

- Infinite-volume limits of quantum lattice systems
- C^* -algebras and their representations
- Superselection theory and braided tensor categories
- Anyons in lattice systems
- Fractional quantum Hall effect and its anyon braiding statistics
- Symmetries in quantum systems
- Computational complexity classes related to games (e.g. RG, MIP)
- Programming: Lean (mathlib contributor), Python

Collaborators and Affiliations

Martin Fraas, Sven Bachman and Yoshiko Ogata - quantum lattice systems with anyons, FQHE
Greg Kuperberg, Sanchayan Dutta, Daniel Tobias - computational complexity